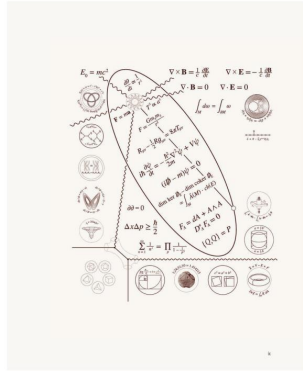


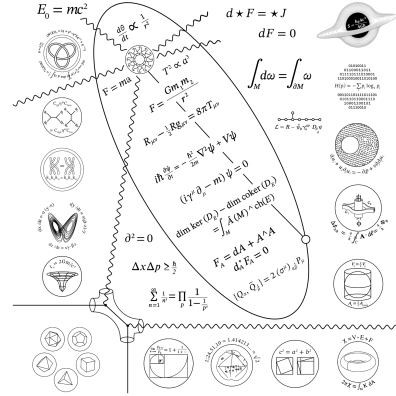
Iconic Wall Change Log

Original Simons Center wall reference → final Mister Consistency fabrication artwork

Revision 5 · August 17, 2026 — the five formulations we chose to restate, with our reasoning (§3); full-page plates of the final and original walls
Rev. 4 (August 16, 2026) legend plaque caught up to the July 31 artwork · Rev. 3 (July 28, 2026) typeset edition · Rev. 2 (July 9, 2026) Maxwell recast in differential forms; Bekenstein–Hawking added



Original reference diagram (catalogue)



Final vector fabrication artwork (July 31, 2026)

Figure 1: Original catalogue wall diagram beside the revised 42×42 vector fabrication artwork (July 31, 2026).

1 Executive summary

- The final Iconic Wall remains a faithful reconstruction of the original Simons Center wall: the central ellipse, peripheral medallions, background equations, and the wall's historical math–physics progression are all preserved.
- The dominant change is production-oriented: the stone-relief photographic reference was converted into monochrome, fabrication-ready vector line art for laser-etched or sandblasted steel panels (42×42 and 32×32 inch, plus a 10×10 inch legend plaque).
- Two entries join the background — Shannon's information theory (I) and the Bekenstein–Hawking entropy (J) — expanding the background key from A–H to A–J, and Maxwell's equations are recast in differential forms. All three are told below.
- The July 31 artwork revision made J pictorial — the formula now sits inscribed on the event horizon of a drawn black hole with accretion disc, top right — and the legend plaque was revised to match (August 2026).
- The legend was rewritten for consistency and specificity: capitalization normalized, long entries shortened, credits tied to removed raster sources dropped, labels made precise. Several formulas were cleaned or clarified — most notably Dirac, Yang–Mills, supersymmetry, Archimedes, boundary-of-boundary notation, and the mass–energy label.
- Revision 5 documents the five formulations we chose to restate, each with its reasoning (§3). Headline: the Jones rewrite, previously logged as a redraw, is a deliberate change of notation ($V_K(t) \rightarrow \langle W(K) \rangle_k$). Full-page plates of both walls follow §3; the detailed inventories now sit in the appendix.

2 The three additions, in brief

B Maxwell's equations, recast. All of electricity, all of magnetism, and light itself, in two short lines: $dF = 0$ and $d\star F = \star J$. When Maxwell first captured these laws in the 1860s they filled twenty equations; vector calculus later compressed them to the famous four; the modern language of differential forms — the same d and $\star$ that appear elsewhere on this wall — folds them into two. Physics got deeper by getting shorter. The four vector-calculus equations of the original wall are honored, not discarded: they are the same physics, written in the wall's own geometric tongue.

I Shannon's information theory. In 1948 a Bell Labs engineer asked a question nobody thought was mathematics: how much surprise does a message carry? Claude Shannon's answer, $H(p) = -\sum_i p_i \log_2 p_i$, measures information in a brand-new unit he named the *bit* — and drew a hard ceiling on how far any message can be compressed and how fast any channel can carry it, forever, regardless of technology. The raw binary carved around the formula is the alphabet Shannon's theorem governs — two symbols, endlessly arranged, now carrying nearly all human communication.

J The Bekenstein–Hawking entropy. One line, four constants, four departments of physics: $S = k_B A c^3 / 4 G \hbar$ carries Newton’s G (gravity), Planck’s $\hbar$ (quantum), Einstein’s c (relativity), and Boltzmann’s k_B (thermodynamics). No other equation makes all four shake hands. It says a black hole has entropy — hidden information — proportional to the *area* of its horizon, not the volume inside: everything a black hole swallows is somehow recorded on its surface. Physicists call the idea holography. The wall’s newest stone, and its best candidate for the physics of the next century.

3 Restated and amended formulations

The final wall does not reproduce every formula exactly as carved: in five places we chose to restate a formulation. This section records each decision: the original as it stands on the wall (quoted from the official catalogue and checked against the carving; where the two disagree, we say so), the form we adopted for the final artwork, and the reasoning behind the choice. Immaterial differences of letterform and spacing are inventoried in the appendix.

3.1 The Jones polynomial

Original — two arcs around the trefoil; the evaluation point sits inside $V_K(\cdot)$:

$$V_K(t) = (t + t^3 - t^4) \left(t^{\frac{1}{2}} + t^{-\frac{1}{2}} \right)$$

$$V_K \left(e^{\frac{2\pi i}{k+2}} \right) = \frac{1}{Z} \int \left(\text{Tr Pexp} \oint_K A \right) e^{\frac{ik}{4\pi} \text{CS}(A)} \mathcal{D}A$$

Final — three formulas; the evaluation point becomes its own statement:

$$\langle W(K) \rangle_k = (q + q^3 - q^4) \left(q^{1/2} + q^{-1/2} \right) \quad q = e^{\frac{2\pi i}{k+2}}$$

$$\langle W(K) \rangle_k = \frac{1}{Z} \int \text{Tr } \mathcal{P} \exp \left(\oint_K A \right) e^{\frac{ik}{4\pi} \text{CS}(A)} \mathcal{D}A$$

Why. One statement, told in order. The bottom arc’s path integral computes a Wilson-loop expectation value, so the final wall names the object honestly — $\langle W(K) \rangle_k$, at level k — instead of borrowing the polynomial’s V_K . Pulling $q = e^{2\pi i/(k+2)}$ out as its own formula states Witten’s dictionary explicitly (evaluate there, and the expectation value is the Jones polynomial), letting the top arc give the trefoil’s value in the standard quantum variable q . Definition, dictionary, value — three arcs, one derivation. Script $\mathcal{P}$ marks path-ordering as an operator rather than a field, and the parentheses now enclose exactly what is exponentiated.

3.2 The boundary of a boundary is zero

$$\partial\partial = 0 \quad \longrightarrow \quad \partial^2 = 0$$

Why. The squared form is the working statement of the theorem: one operator, composed with itself, annihilating everything — which is how homology actually wields it, and how the rest of the final wall already speaks (the exterior calculus of its Stokes and Maxwell items, where $d^2 = 0$ is the mirror identity). $\partial\partial$ juxtaposes two symbols; ∂^2 asserts the identity. Same truth, sharpened to its modern form.

3.3 The Dirac equation

$$(i\cancel{D} - m)\psi = 0 \quad \longrightarrow \quad (i\gamma^\mu \partial_\mu - m)\psi = 0$$

Why. The slash is shorthand a reader must already own, and the capital $\cancel{D}$ conventionally smuggles in a gauge field — leaving the carving suspended between the free equation and the coupled one. The final wall writes the mechanism in full: gamma matrices contracting the spacetime derivative, the Clifford square root of the wave operator. That is Dirac’s 1928 equation with nothing left to convention; coupling then enters by the visible substitution $\partial \rightarrow D$, not by silent capitalization.

3.4 The Yang–Mills equations

$$F = dA + A \wedge A \quad D_A^* F_A = 0 \quad \longrightarrow \quad F_A = dA + A \wedge A \quad d_A^* F_A = 0$$

The stone carves the first F bare; the catalogue’s typeset panel silently adds the subscript (F_A). The final wall follows the catalogue.

Why. Two consistency repairs. The stone names the curvature F in line one and F_A in line two; the final wall subscripts both, so the object’s dependence on the connection A is explicit and the pair closes on itself — the F_A annihilated in line two is the F_A defined in line one. And the operator family is unified: line one builds curvature from the lowercase exterior derivative d ; line two applies d_A^* , the adjoint of its covariant extension d_A — staying in the lowercase calculus where the original’s uppercase D_A^* switched alphabets mid-pair. One connection, one calculus, one case.

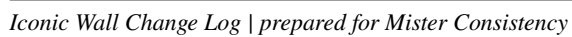
3.5 Supersymmetry

$$\{Q, Q\} = P \quad \longrightarrow \quad \{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2(\sigma^\mu)_{\alpha\dot{\beta}} P_\mu$$

Why. The stone carves the catalogue’s admitted shorthand — “an equation for the operator Q ” — with indices, conjugation, and normalization all suppressed. The final wall carves the theorem: in two-component form, supercharge and conjugate supercharge close on the momentum through the sigma matrices. It is the exact statement the slogan gestures at — the square of a supersymmetry is a translation — now able to survive a reader who checks the indices.

July 31, 2026 vector fabrication artwork, 42 × 42

July 31, 2026 vector fabrication artwork, 42 × 42



Simons Center for Geometry and Physics, Stony Brook — carved limestone, dedicated May 8, 2015



Appendix A Summary changes by category

Area	Original	Final	Change made
Overall wall	Carved/sculpted wall documentation and catalogue diagram.	Rebuilt as clean, square, fabrication-ready vector line art.	Major production change: from photographic/relief reference to flat black vector art.
Overall format	Original installation is tall: $279 \times 240 \times 2$ inches; the catalogue describes a $23 \text{ ft} \times 20 \text{ ft}$ wall.	Square 42×42 and 32×32 inch panels, plus a 10×10 inch legend plaque.	Scale and aspect ratio changed for steel-panel fabrication.
Color / material	Stone relief, shadows, tonal texture, catalogue teal headings.	Pure black line art; no gradients, raster images, or shadows.	All tonal and material cues removed for engraving clarity.
All medallions	Mix of diagram line work, relief, and several photo-based medallions.	All medallions redrawn as consistent vector strokes.	Unifies style; improves reproducibility at reduced sizes.
Central ellipse	Large diagonal ellipse with sun, planet path, and equations 1–10.	Retained; line weights, spacing, and surrounding formulas cleaned for the square layout.	Core composition preserved.
Photon lines	Sculptural relief-like wavy lines crossing the wall.	Converted to a high-contrast vector stroke pattern.	Fabrication readability.
Background equations	Four vector-calculus Maxwell equations; background items A–H.	Maxwell recast as $dF = 0$, $d\star F = \star J$; Shannon (I) and Bekenstein–Hawking (J) blocks added.	Notation modernized; two new background entries.
Legend system	Key uses I–XIII, 1–10, and background A–H.	Key uses I–XIII, 1–10, and background A–J; plaque revised August 2026 to mirror the July 31 layout, with J drawn as its black-hole silhouette.	The two additions required two new background key entries; locator updated to the current wall.

Appendix B Legend changes: original → final

The final legend plaque is not a verbatim copy of the catalogue legend: it is a rewritten key for a new fabrication object. The numbering system remains; the wording is normalized and two background items are added.

Key	Original wording	Final wording	Change
Heading	Key to Equations and Diagrams	Key to equations and diagrams (I–XIII)	Lowercase style normalized; numbering range made explicit.
I	The Jones Polynomial	The Jones polynomial	Capitalization normalized.
II	Associativity in Quantum Field Theory	Associativity in quantum field theory	Capitalization normalized.
III	The Yang-Baxter Equation	The Yang–Baxter equation	Capitalization normalized; dash treatment standardized.
IV	The Lorenz Attractor (Image: Scott Sutherland)	The Lorenz attractor	Capitalization normalized; image credit removed because the final is redrawn line art.

Key	Original wording	Final wording	Change
V	The Schwarzschild Black Hole	The Schwarzschild black hole	Capitalization normalized; spelling verified.
VI	The Five Platonic Solids	The five Platonic solids	Capitalization normalized.
VII	The Golden Ratio	The golden ratio	Capitalization normalized.
VIII	The Babylonian tablet YBC 7289 (Image: Bill Casselman)	The Babylonian tablet YBC 7289	Raster-photo attribution removed; the final tablet is vector line art.
IX	The Pythagorean Theorem	The Pythagorean theorem	Capitalization normalized.
X	The Gauss-Bonnet Theorem	The Gauss–Bonnet theorem	Capitalization normalized; dash standardized.
XI	Archimedes’ calculation of the volume and area of the sphere	Archimedes’ sphere and cylinder	Long label shortened to the medalion’s visual subject.
XII	The Aharonov-Bohm Effect	The Aharonov–Bohm effect	Capitalization normalized; dash standardized.
XIII	The Navier-Stokes Equations (Image: J.D. Kim, AMS, Stony Brook)	The Navier–Stokes equations	Capitalization normalized; image credit removed (redrawn as line art).
Ellipse heading	In the Ellipse	Within the ellipse (1–10)	Heading made natural; numbering range explicit.
Ellipse note	(Kepler’s 1st law represented by star, ellipse, planet)	(Kepler’s 1st law represented by star, ellipse, planet)	Kept; surrounding legend typography cleaned.
2	Newton’s Second Law of Motion	Newton’s 2nd law of motion	Numeral style harmonized with Kepler’s 1st/2nd/3rd laws.
Background heading	On the Background	On the background (A–J)	Capitalization normalized; expanded from A–H to A–J.
A	$E=mc^2$	Einstein’s rest mass–energy equivalence	Formula-only label replaced with a descriptive title.
D	Supergravity	Minimal supergravity Lagrangian	Made specific to the displayed formula.
H	Feynman Diagrams	Feynman and string interaction “pair-of-pants”	Expanded to reflect the combined particle/string drawing.
I	<i>No original entry</i>	Shannon’s information theory	New background item added to the final artwork and legend.
J	<i>No original entry</i>	Bekenstein–Hawking entropy	New background item added in the July 2026 revision; wording tightened and the locator glyph redrawn as the black hole with accretion disc (August 2026).

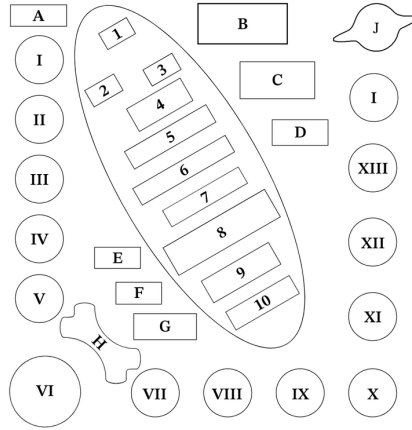
Key to equations and diagrams (I–XIII)

- I The Jones polynomial
- II Associativity in quantum field theory
- III The Yang–Baxter equation
- IV The Lorenz attractor
- V The Schwarzschild black hole
- VI The five Platonic solids
- VII The golden ratio
- VIII The Babylonian tablet YBC 7289
- IX The Pythagorean theorem
- X The Gauss–Bonnet theorem
- XI Archimedes’ sphere and cylinder
- XII The Aharonov–Bohm effect
- XIII The Navier–Stokes equations

Within the ellipse (1–10)

(Kepler’s 1st law represented by star, ellipse, planet)

- 1 Kepler’s 2nd law
- 2 Newton’s 2nd law of motion
- 3 Kepler’s 3rd law
- 4 Newton’s law of gravitation
- 5 General relativity
- 6 Schrödinger’s equation
- 7 The Dirac equation
- 8 The Atiyah–Singer index theorem
- 9 The Yang–Mills equations
- 10 Supersymmetry



On the background (A–J)

- A Einstein’s rest mass–energy equivalence
- B Maxwell’s equations
- C Stokes’ theorem
- D Minimal supergravity Lagrangian
- E The boundary of a boundary is zero
- F Heisenberg’s uncertainty principle
- G The Riemann zeta function
- H Feynman and string interaction “pair-of-pants”
- I Shannon’s information theory
- J Bekenstein–Hawking entropy

Figure 2: The 10 × 10 legend plaque, August 2026 revision. The locator mirrors the July 31 wall: the Bekenstein–Hawking entry (J) appears as its black-hole silhouette at the top right, with Shannon (I) and Navier–Stokes (XIII) following down the right edge.

Appendix C Element-by-element changes

The following tables list every named item in the original key, the ten ellipse items, and every background item. “Retained” does not mean visually identical; in nearly every case the final version was redrawn or retypeset as consistent black vector art.

C.1 Peripheral diagrams and medallions (I–XIII)

Item	Original reference	Final version	Change type
I Jones polyno- mial	Trefoil knot ringed by $V_K(t)$ and Witten’s Chern–Simons path integral.	Notation rewritten to Wilson-loop form $\langle W(K) \rangle_k$ with $q = e^{2\pi i/(k+2)}$ stated separately; redrawn as vector line art (§3.1).	Notation change, redraw.
II Associativity in QFT	Tree-level associativity diagram.	Retained; redrawn.	Redraw, typography.
III Yang–Baxter equation	Braiding equality diagram with R -matrix expression.	Retained; redrawn.	Redraw, typography.
IV Lorenz attrac- tor	Butterfly image credited to Scott Sutherland.	Redrawn as vector attractor; the three equations are symbol-for-symbol unchanged (letterforms and spacing only); credit removed.	Photo to vector.

Item	Original reference	Final version	Change type
V Schwarzschild black hole	Embedding-diagram medallion.	Retained; redrawn with clean vector curves.	Redraw, spelling check.
VI Five Platonic solids	Five separate solid diagrams.	Retained; redrawn with consistent outlines.	Redraw.
VII Golden ratio	Golden rectangle/spiral with limit and continued fraction.	Retained; redrawn.	Redraw, typography.
VIII Babylonian tablet	Tablet photograph credited to Bill Casselman.	Simplified line-art tablet medallion; credit removed.	Photo to vector.
IX Pythagorean theorem	Wordless-proof diagrams.	Retained; redrawn.	Redraw, typography.
X Gauss–Bonnet theorem	Torus medallion and formula.	Retained; cleaner torus and formula placement.	Redraw.
XI Archimedes	“Calculation of the volume and area of the sphere”; formula shown generically.	Label shortened; formula clarified as the sphere–cylinder relationship, $V_s = \frac{2}{3}V_c$, $A_s = \frac{2}{3}A_{c,tot}$.	Label, notation.
XII Aharonov–Bohm effect	Magnetic flux-tube and path diagram.	Retained; formula and paths cleaned.	Redraw, typography.
XIII Navier–Stokes	Flow image credited to J.D. Kim / AMS / Stony Brook.	Vectorized line-art vortex; credit removed; repositioned to the right edge, below Shannon’s block (July 31 layout).	Image to vector.

C.2 Ellipse items (Kepler + 1–10)

Item	Original	Final	Change type
Kepler’s 1st law	Star, ellipse, planet represented compositionally.	Retained; ellipse and sun lines cleaned in the square layout.	Redraw, layout.
1 Kepler’s 2nd law	$d\theta/dt \propto 1/r^2$.	Retained; retypeset and repositioned.	Typographic cleanup.
2 Newton’s 2nd law	$F = ma$.	Retained; legend “Second” $\rightarrow$ “2nd”.	Legend style.
3 Kepler’s 3rd law	$T^2 \propto a^3$.	Retained; retypeset.	Typographic cleanup.
4 Newton’s gravitation	$F = Gm_1m_2/r^2$.	Retained; retypeset.	Typographic cleanup.
5 General relativity	Einstein field equation.	Retained; retypeset with cleaner spacing.	Typographic cleanup.
6 Schrödinger equation	Nonrelativistic Schrödinger equation.	Retained; clearer $\hbar$, Laplacian, and potential notation.	Typographic cleanup.
7 Dirac equation	$(i\not{D} - m)\psi = 0$ (compact slashed form).	$(i\gamma^\mu\partial_\mu - m)\psi = 0$ (explicit form, §3.3).	Notation change.
8 Atiyah–Singer	Index formula.	Retained; retypeset.	Typographic cleanup.
9 Yang–Mills	$F = dA + A \wedge A$; $D_A^*F_A = 0$ (stone carves the first F bare; the catalogue typesets F_A).	$F_A = dA + A \wedge A$; $d_A^*F_A = 0$ (§3.4).	Subscripts, operator case.

Item	Original	Final	Change type
10 Supersymmetry	$\{Q, Q\} = P$ schematic.	$\{Q_\alpha, \bar{Q}_\beta\} = 2(\sigma^\mu)_{\alpha\beta} P_\mu$ (§3.5).	Formula made explicit.

C.3 Background items (A–J)

Item	Original	Final	Change type
A Mass–energy	Background label “E=mc ² ”.	Legend reads “Einstein’s rest mass–energy equivalence”; artwork uses $E_0 = mc^2$.	Name, rest-energy notation.
B Maxwell’s equations	Four equations: $\nabla \times B = \frac{1}{c} \partial E / \partial t$, $\nabla \times E = -\frac{1}{c} \partial B / \partial t$, $\nabla \cdot B = 0$, $\nabla \cdot E = 0$.	Recast as $dF = 0$ and $d\star F = \star J$ in one compact block.	Vector calculus → differential forms (July 2026).
C Stokes’ theorem	Integral formula.	Retained; retypeset.	Typographic cleanup.
D Supergravity	Generic “Supergravity” label and formula.	Legend specifies “Minimal supergravity Lagrangian”; formula redrawn beside the Dynkin diagram.	Label specificity, redraw.
E Boundary of a boundary	$\partial\partial = 0$.	$\partial^2 = 0$ (§3.2).	Equivalent, more compact.
F Heisenberg uncertainty	$\Delta x \Delta p \geq \hbar/2$.	Retained; retypeset.	Typographic cleanup.
G Riemann zeta	Euler product–sum relationship, product on the left: $\prod_p \frac{1}{1-1/p^s} = \sum_{n=1}^\infty \frac{1}{n^s}$.	Sides exchanged — the artwork reads $\sum_{n=1}^\infty \frac{1}{n^s} = \prod_p \frac{1}{1-1/p^s}$; retypeset.	Order, typography.
H Feynman diagrams	Particle diagram and string–interaction detail.	Legend expanded to “Feynman and string interaction pair-of-pants”.	Label expanded.
I Shannon	<i>No original entry.</i>	New block: $H(p) = -\sum_i p_i \log_2 p_i$ with carved binary strings.	New content.
J Bekenstein–Hawking	<i>No original entry.</i>	New medallion at the top right of the wall: a black hole ringed by its accretion disc, with $S = k_B A c^3 / 4G\hbar$ inscribed on the event horizon — gravity, quantum mechanics, thermodynamics, and information in one line.	New content (July 2026); pictorial black-hole medallion (July 31, 2026).